

AIIMS CRE — OT / Anaesthesia Technician

Comprehensive Practice Question Bank — 100 Original MCQs with Explanations

Covers OT Technology • Anaesthesia Equipment • Pharmacology • Monitoring & Emergency • Surgical Procedures
Original practice questions — not leaked items. Cross-check high-stakes facts with standard textbooks before the exam.

Instructions:

- Each question has four options (A–D). Select the single best answer.
- Answers and explanations follow each question for self-assessment.
- Total: 100 Questions | Five Sections (A–E) | 20 questions per section.
- Recommended time: 90–120 minutes for exam simulation.

Section A — OT Technology, Sterilization & Surgical Instruments (Q1–20)

Q1. The 'positive pressure' maintained inside a standard operating theatre (relative to the corridor) serves primarily to:

- A) Conserve electricity
- B) Prevent ingress of contaminated air from outside
- C) Maintain higher humidity
- D) Reduce anaesthetic gas concentration

✓ **Answer: B**

Explanation: Positive-pressure ventilation in the OT ensures that airflow is always outward, pushing potentially contaminated corridor air away from the sterile field and reducing surgical-site infection risk.

Q2. Which sterilization method is used for fibre-optic endoscopes because it operates at low temperatures and uses hydrogen peroxide plasma?

- A) Autoclave (steam)
- B) Dry heat oven
- C) STERRAD (Hydrogen Peroxide Plasma)
- D) Boiling water

✓ **Answer: C**

Explanation: STERRAD systems vaporize hydrogen peroxide and generate a low-temperature plasma, achieving sterilization at <50°C — ideal for heat-labile scopes and cameras that would be damaged by steam or ETO.

Q3. Flash sterilization (Immediate-Use Steam Sterilization, IUSS) at 134°C/30 psi for unwrapped instruments is intended for:

- A) Routine overnight sterilization
- B) Emergency re-sterilization of a dropped instrument needed immediately
- C) Sterilizing plastic IV tubing
- D) Sterilizing powered surgical drills

✓ **Answer: B**

Explanation: IUSS is a rapid cycle (3–4 min at 134°C) reserved for urgent situations such as an instrument dropped in the sterile field; it is not recommended for routine use because no sterility-maintenance packaging is applied.

Q4. The chemical indicator strip placed INSIDE a sterilization pack (Type 5 integrating indicator) changes colour when:

- A) The pack is exposed to any heat
- B) All three critical parameters — time, temperature, and steam — are met
- C) The outer of the pack is wet
- D) The autoclave door is closed

✓ **Answer: B**

Explanation: Type 5 integrating indicators respond to all critical sterilization variables (time + temperature + steam), providing stronger proof of efficacy than simple process indicators that only confirm heat exposure.

Q5. The minimum safe immersion time in 2% activated glutaraldehyde to achieve HIGH-LEVEL DISINFECTION (not full sterilization) is:

- A) 1 minute
- B) 5 minutes
- C) 20–30 minutes
- D) 10 hours

✓ **Answer: C**

Explanation: High-level disinfection (eliminating all organisms except large numbers of bacterial spores) requires 20–30 minutes contact with 2% glutaraldehyde; 10 hours achieves full sterilization.

Q6. Formaldehyde fumigation of the OT is performed at a concentration of:

- A) 0.5 g/m³
- B) 1 g/m³
- C) 4–8 g/m³
- D) 20 g/m³

✓ **Answer: C**

Explanation: Standard OT terminal fumigation uses formalin at 4–8 g formaldehyde per m³ of room volume, generated by heating formalin tablets or mixing with potassium permanganate; the room is sealed for 6–12 hours.

Q7. Which colour-coded bag is used for cytotoxic/chemotherapy waste according to India's Biomedical Waste Management Rules 2016?

- A) Red bag
- B) Yellow bag
- C) Black bag
- D) Blue bag

✓ **Answer: B**

Explanation: Cytotoxic and pharmaceutical (including chemotherapy) waste is categorised as Category 1 biomedical waste in India and is collected in yellow bags/containers for incineration.

Q8. In OT instrument layout, the 'Mayo stand' is best described as:

- A) A self-retaining abdominal retractor
- B) A mobile stainless-steel stand positioned over the patient to hold immediately-needed instruments
- C) A fixed wall-mounted instrument rail
- D) An anaesthesia trolley

✓ **Answer: B**

Explanation: The Mayo stand is a height-adjustable, wheeled over-patient table covered with a sterile drape on which the scrub nurse/technician arranges instruments needed immediately during the procedure.

Q9. The Doyen retractor is a curved, flat-bladed instrument primarily used to retract:

- A) The liver during cholecystectomy
- B) The bladder/uterus during pelvic surgery
- C) The lung during thoracotomy
- D) The skin edges only

✓ **Answer: B**

Explanation: The Doyen retractor has a broad, curved blade designed to retract the bladder and uterus during lower abdominal and pelvic procedures such as caesarean section and hysterectomy.

Q10. Gillies tissue forceps differ from standard toothed forceps in that they:

- A) Have no teeth
- B) Have interdigitating fine teeth (1×2) causing less crushing than Bonney forceps
- C) Are designed only for bone
- D) Are self-locking

✓ **Answer: B**

Explanation: Gillies (Adson) tissue forceps have finely interdigitating teeth producing a firm but relatively atraumatic grip on skin and subcutaneous tissue — less crushing than the heavier Bonney or Russian forceps.

Q11. Which needle type, described as a 'cutting' needle, is most suitable for skin closure?

- A) Tapered (round-bodied) needle
- B) Reverse-cutting needle
- C) Blunt needle
- D) Keith straight needle

✓ **Answer: B**

Explanation: Reverse-cutting needles have the cutting edge on the outer curve, reducing the risk of the suture cutting through tissue compared to conventional cutting needles — making them the standard choice for skin.

Q12. Polydioxanone (PDS) is a synthetic absorbable suture that retains 50% of its tensile strength for approximately:

- A) 3–5 days
- B) 14 days
- C) 4–6 weeks
- D) 6 months

✓ **Answer: C**

Explanation: PDS undergoes slow hydrolysis, retaining ~70% strength at 2 weeks and ~50% at 4–6 weeks, making it suitable for slow-healing tissues such as abdominal wall fascia and paediatric cardiac repair.

Q13. In bipolar electrosurgery (compared to monopolar), the major safety advantage is that:

- A) Higher current is used
- B) Current passes only between the two tines of the forceps, not through the patient's entire body
- C) A grounding pad is always required
- D) It can cut tissue more effectively

✓ **Answer: B**

Explanation: Bipolar diathermy confines current between the two forcep tips, eliminating risk of inadvertent burns at distant sites (e.g., cardiac pacemaker, grounding-pad area) and removing the need for a return electrode.

Q14. The term 'aseptic technique' means:

- A) Killing all organisms on living tissue with antiseptics
- B) Preventing contamination of sterile fields, instruments, and wounds by maintaining a bacteria-free environment
- C) Sterilization of reusable instruments only
- D) Using antibiotics before surgery

✓ **Answer: B**

Explanation: Aseptic technique encompasses all practices — sterile gowning, gloving, draping, and instrument handling — aimed at preventing introduction of microorganisms into the surgical site.

Q15. During a surgical hand scrub, which area is scrubbed FIRST following the counted-stroke method?

- A) Forearm
- B) Palm
- C) Fingernails and fingertips
- D) Dorsum of hand

✓ **Answer: C**

Explanation: The counted-stroke scrub proceeds in a defined order: nails/fingertips first (where most bacteria reside), then each finger surface, then palm, dorsum, and finally the forearm — always brushing from most-contaminated to least.

Q16. The 'transfer zone' in the OT complex is used specifically to:

- A) Store sterilized instruments
- B) Transfer patients from ward trolley to OT trolley, maintaining the clean/dirty boundary
- C) Perform pre-operative anaesthesia induction
- D) Dispose of soiled linen

✓ **Answer: B**

Explanation: The transfer zone (protective zone boundary) contains separate trolleys on each side of a marked line; the patient is moved across this line to an OT-dedicated trolley, preventing ward-contaminated equipment from entering the clean zone.

Q17. Which instrument is used specifically to dilate the cervical os during uterine procedures (e.g., D&C;)?

- A) Hegar dilators
- B) Pratt dilators only
- C) Trocar and cannula
- D) Veress needle

✓ **Answer: A**

Explanation: Hegar dilators are graduated metallic rods used in sequential sizes to gently dilate the cervical canal before intrauterine procedures; Pratt dilators are a similar set with a tapered double end.

Q18. In an OT, 'traffic flow' is controlled primarily to:

- A) Speed up surgical procedures
- B) Reduce the number of airborne particles and microorganisms generated by movement in and out of the sterile zone
- C) Facilitate patient transport only
- D) Comply only with fire regulations

✓ **Answer: B**

Explanation: Each person entering or moving within the OT sheds skin squames and microorganisms; minimising unnecessary movement ('traffic') is one of the most effective measures to keep airborne bacterial counts low.

Q19. The 'instrument count' discrepancy at skin closure is most appropriately managed by:

- A) Closing the wound and recounting outside
- B) Performing intraoperative X-ray/fluoroscopy of the wound before closure
- C) Ignoring it if the surgeon is confident
- D) Documenting the discrepancy and proceeding

✓ **Answer: B**

Explanation: An unresolved count discrepancy is a patient-safety emergency; intraoperative imaging of the body cavity is performed before final closure to identify any retained item, as per universal surgical safety protocols.

Q20. The surgical helmet ('space suit') with a personal air supply is routinely worn by the surgical team during:

- A) Appendectomy
- B) Total joint replacement (arthroplasty) in ultra-clean air environments
- C) Thyroidectomy
- D) Cataract surgery

✓ **Answer: B**

Explanation: Body-exhaust suits with integrated filtered air reduce the shedding of skin organisms into the ultra-clean (laminar-flow) OT environment during high-risk implant procedures like hip and knee replacement.

Section B — Anaesthesia Equipment & Airway Management (Q21–40)

Q21. In a circle breathing system, soda lime absorbs CO₂ by a chemical reaction whose by-products are:

- A) Nitrogen gas and water
- B) Water, heat, and calcium carbonate
- C) Oxygen and argon
- D) Hydrogen and carbon monoxide

✓ **Answer: B**

Explanation: $\text{CO}_2 + \text{Ca(OH)}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O} + \text{heat}$. The reaction produces water and heat (which warms and humidifies inspired gas) and calcium carbonate. Colour indicator dyes (e.g., ethyl violet) turn from white to purple/blue as the soda lime is exhausted.

Q22. The Adjustable Pressure-Limiting (APL) valve in a breathing circuit is also called the:

- A) PEEP valve
- B) Expiratory valve or 'pop-off' valve
- C) Inspiratory check valve
- D) Oxygen flush valve

✓ **Answer: B**

Explanation: The APL (pop-off) valve vents excess gas from the circuit during expiration; when partially closed (e.g., during spontaneous ventilation with a bag) it allows the operator to control the degree of pressure build-up.

Q23. The oxygen flush button/lever on an anaesthesia machine delivers oxygen directly to the common gas outlet at a flow of approximately:

- A) 2 L/min
- B) 15 L/min
- C) 35–75 L/min
- D) 100 L/min

✓ **Answer: C**

Explanation: The emergency oxygen flush delivers O₂ at 35–75 L/min, bypassing the flowmeters and vaporizer — useful for rapidly flushing the circuit, but it can barotrauma a patient if used carelessly during mechanical ventilation.

Q24. Which pipeline colour carries NITROUS OXIDE in the Indian/UK medical gas pipeline system?

- A) Blue
- B) French blue (royal blue)
- C) Grey
- D) White

✓ **Answer: B**

Explanation: In the Indian (IS 3933) and UK (BS EN 737) systems, nitrous oxide pipelines are identified by French blue (royal blue) — distinct from oxygen (white in UK; black-and-white in some older Indian systems).

Q25. Variable bypass vaporizers (e.g., TEC series) are calibrated for a specific volatile agent. Using the wrong agent in a vaporizer is dangerous primarily because agents differ in their:

- A) Molecular weight only
- B) Saturated vapour pressure (SVP), leading to incorrect delivered concentrations
- C) Colour only
- D) Boiling points irrelevant to output

✓ **Answer: B**

Explanation: Each agent has a unique SVP; an agent with a higher SVP in a calibrated vaporizer would deliver dangerously high concentrations. Agent-specific filling adaptors (QUIK-Fill/Selectatec) are safety features to prevent misfilling.

Q26. A 'Guedel' (oropharyngeal) airway is contraindicated in:

- A) Unconscious patients without a gag reflex
- B) Semiconscious patients who retain a gag reflex (risk of vomiting/laryngospasm)
- C) Patients with dentures removed
- D) Patients in prone position

✓ **Answer: B**

Explanation: The Guedel airway may trigger vomiting or laryngospasm in any patient who retains a gag reflex; it is only safe in deeply unconscious patients. A nasopharyngeal airway is better tolerated in obtunded but not fully unconscious patients.

Q27. Sellick's manoeuvre (cricoid pressure) applied during rapid sequence induction aims to:

- A) Facilitate laryngoscopy by displacing the larynx posteriorly
- B) Compress the oesophagus against the vertebral body to reduce passive regurgitation risk
- C) Open the mouth
- D) Stimulate breathing

✓ **Answer: B**

Explanation: Pressing the cricoid cartilage posteriorly occludes the oesophagus against C5–C6 vertebrae, reducing risk of silent regurgitation of gastric contents during the apnoeic interval of RSI, though its routine use is debated.

Q28. Correct endotracheal tube placement is most reliably confirmed intraoperatively by:

- A) Auscultation of breath sounds only
- B) Continuous capnography showing sustained CO₂ waveform
- C) Fogging of the ETT
- D) Observation of chest rise alone

✓ **Answer: B**

Explanation: Sustained end-tidal CO₂ waveform on capnography is the gold standard — oesophageal intubation may briefly show a small CO₂ trace from swallowed air but it rapidly disappears, unlike the consistent square wave of correct tracheal placement.

Q29. A 'bougie' (endotracheal tube introducer/gum elastic bougie) is used to:

- A) Suction secretions from the airway
- B) Aid intubation when the larynx is not fully visible by providing a rail over which the ETT is railroaded
- C) Measure tracheal diameter
- D) Deliver oxygen only

✓ **Answer: B**

Explanation: The bougie's angled tip is felt 'clicking' over tracheal rings (confirming tracheal placement) and the ETT is advanced over it into the trachea — a key tool for grades 3 and 4 Cormack-Lehane views.

Q30. The Cormack-Lehane Grade III view during direct laryngoscopy means:

- A) Full view of the glottis
- B) Partial view of the glottis
- C) Only the epiglottis is visible
- D) Neither glottis nor epiglottis visible

✓ **Answer: C**

Explanation: Cormack-Lehane Grade III = only the epiglottis is visible (no glottic structures seen); Grade IV = neither epiglottis nor glottis is visible. Grades III–IV are associated with difficult or failed intubation.

Q31. The Tuohy needle used for epidural anaesthesia has a curved (Huber) tip designed to:

- A) Cut through the ligamentum flavum rapidly
- B) Direct the epidural catheter cephalad rather than puncture the dura
- C) Withdraw dural fibres
- D) Measure epidural space depth

✓ **Answer: B**

Explanation: The Huber tip deflects the threading catheter laterally and cephalad into the epidural space, reducing the risk of inadvertent dural puncture while also discouraging the catheter from threading into an epidural vein.

Q32. Accidental dural puncture during epidural placement most commonly causes:

- A) Permanent paralysis
- B) Postdural puncture headache (PDPH), characteristically worse when upright
- C) Hypertension
- D) Pneumothorax

✓ **Answer: B**

Explanation: CSF leakage through the dural hole lowers intracranial pressure, causing a postural headache (worse sitting/standing, relieved by lying flat) that may require an epidural blood patch if conservative measures fail.

Q33. Which inhalational agent is known to have the HIGHEST MAC value (least potent)?

- A) Halothane (MAC ~0.75%)
- B) Isoflurane (MAC ~1.2%)
- C) Sevoflurane (MAC ~2%)
- D) Desflurane (MAC ~6–7%)

✓ **Answer: D**

Explanation: MAC and potency are inversely related. Desflurane has the highest MAC (~6–7%), meaning the largest concentration is needed for effect — making it the least potent. Halothane's low MAC (~0.75%) makes it the most potent of the listed agents.

Q34. Total Intravenous Anaesthesia (TIVA) using a target-controlled infusion (TCI) pump most commonly uses which drug combination?

- A) Ketamine and midazolam
- B) Propofol (hypnotic) and remifentanil (opioid)
- C) Thiopentone and morphine
- D) Etomidate and fentanyl

✓ **Answer: B**

Explanation: Propofol-remifentanil TIVA delivers titrated hypnosis and analgesia without volatile agents; remifentanil's ultra-short action (esterase metabolism) allows rapid adjustment of analgesic depth during surgery.

Q35. A 'DISS' (Diameter-Index Safety System) connection, as opposed to a pin-index system, is used for:

- A) Small high-pressure cylinder connections
- B) Low-pressure pipeline hose connections at the anaesthesia machine wall outlet
- C) Vaporizer filling ports
- D) Oxygen cylinder regulators only

✓ **Answer: B**

Explanation: DISS connectors are non-interchangeable threaded fittings used for the low-pressure (pipeline) wall outlets and machine hose connections, preventing wrong-gas attachment in the piped gas system — a different safety system from the pin-index used for cylinders.

Q36. The 'Mapleson E' circuit (Ayre's T-piece) is most commonly used for:

- A) Adults under controlled ventilation
- B) Neonates and small children during spontaneous ventilation
- C) Patients with high CO₂ retention
- D) Hyperbaric oxygen therapy

✓ **Answer: B**

Explanation: Ayre's T-piece has no valves or reservoir bag, offering minimal resistance and dead space — critical for neonates and small children whose small tidal volumes cannot overcome valve resistance.

Q37. In which situation is a double-lumen endobronchial tube (DLT) specifically indicated?

- A) Routine appendectomy
- B) One-lung ventilation for thoracic surgery (e.g., pneumonectomy, oesophagectomy)
- C) Awake fiberoptic intubation only
- D) Paediatric airway management

✓ **Answer: B**

Explanation: DLTs (e.g., Robertshaw) allow independent ventilation or isolation of each lung, essential during thoracic procedures where the operated lung must be deflated for surgical exposure while the contralateral lung maintains gas exchange.

Q38. The 'Heat and Moisture Exchanger' (HME, artificial nose) placed in the breathing circuit:

- A) Cools inspired gases
- B) Passively conserves heat and moisture from exhaled gas to warm and humidify subsequent inspired gas
- C) Absorbs volatile anaesthetic agents
- D) Filters CO₂

✓ **Answer: B**

Explanation: HMEs trap heat and water from exhaled gas in a hygroscopic element and return it to the patient on the next inspiration, preventing airway drying during prolonged intubation — important for both OT and ICU patients.

Q39. An anaesthesia machine check reveals that the oxygen flowmeter tube is cracked. The MOST immediate safety risk is:

- A) Hypoxic gas mixture reaching the patient due to unmetered oxygen leakage
- B) Excessive humidity
- C) Vaporizer malfunction
- D) APL valve stuck open

✓ **Answer: A**

Explanation: A cracked flowmeter tube causes an inaccurate reading; the actual flow will differ from the indicated flow, potentially allowing a hypoxic (N₂O-dominant) mixture without the operator realising — the machine should be taken out of service immediately.

Q40. The minimum fresh gas flow required to prevent CO₂ rebreathing in a Mapleson D circuit during controlled ventilation is approximately:

- A) 0.5 × minute ventilation
- B) 1 × alveolar ventilation (approximately 70 mL/kg/min)
- C) 2 × minute ventilation
- D) 4 × minute ventilation

✓ **Answer: C**

Explanation: For controlled ventilation with a Mapleson D, a fresh gas flow of approximately 2× the patient's minute ventilation (~150–200 mL/kg/min) prevents CO₂ rebreathing; the Bain (coaxial Mapleson D) shares this requirement.

Section C — Anaesthesia Drugs & Pharmacology (Q41–60)

Q41. The induction agent KETAMINE characteristically produces all of the following EXCEPT:

- A) Dissociative anaesthesia
- B) Bronchodilation (useful in asthma)
- C) Increased secretions (requiring antisialagogue premedication)
- D) Significant respiratory depression and apnoea at standard doses

✓ **Answer: D**

Explanation: Ketamine generally PRESERVES spontaneous respiration and airway reflexes at standard induction doses — a key advantage. It increases secretions, produces bronchodilation (useful in asthmatic patients), and causes dissociative anaesthesia with vivid emergence phenomena.

Q42. Midazolam's unique advantage as a premedication over other benzodiazepines is primarily its:

- A) Long duration of action
- B) Water solubility at acidic pH (less pain on injection) and short half-life with anterograde amnesia
- C) Analgesic properties
- D) Muscle-relaxant effect only

✓ **Answer: B**

Explanation: Midazolam is water-soluble at acidic pH in its commercial preparation (less painful IM/IV injection), becomes lipid-soluble at physiological pH, has a short half-life, and produces reliable anterograde amnesia — making it the most widely used premedication benzodiazepine.

Q43. Morphine's release of histamine from mast cells can clinically cause:

- A) Hypertension and bradycardia
- B) Urticaria, hypotension, and bronchospasm
- C) Increased gastric acid secretion
- D) Muscle rigidity only

✓ **Answer: B**

Explanation: Histamine release from morphine can cause urticaria (itching/wheel), vasodilatation with hypotension, and bronchospasm — especially problematic in asthmatics. Synthetic opioids like fentanyl and remifentanyl do not cause histamine release.

Q44. Naloxone, the opioid reversal agent, has a shorter duration than most opioids. This means:

- A) A single dose is always sufficient
- B) Re-narcotisation (return of opioid effect) can occur; repeat dosing or an infusion may be needed
- C) It should only be given subcutaneously
- D) It has no side effects

✓ **Answer: B**

Explanation: Naloxone's half-life is ~60–90 minutes, shorter than most opioids; patients must be monitored for return of respiratory depression after initial reversal, and re-dosing or a continuous infusion may be required.

Q45. Glycopyrrolate, unlike atropine, does NOT cross the blood-brain barrier. This means it:

- A) Is less effective at drying secretions
- B) Does not cause central anticholinergic effects (confusion, sedation) seen with atropine
- C) Cannot be given intravenously
- D) Has a shorter duration of action

✓ **Answer: B**

Explanation: Because glycopyrrolate is a quaternary ammonium compound (charged, water-soluble), it does not enter the CNS and lacks atropine's central side effects of excitation, confusion, and tachycardia; it is also longer-acting.

Q46. Ondansetron prevents postoperative nausea and vomiting (PONV) by acting as a:

- A) Dopamine D2 receptor antagonist
- B) 5-HT₃ (serotonin) receptor antagonist
- C) H₁ histamine receptor antagonist
- D) Muscarinic receptor antagonist

✓ **Answer: B**

Explanation: Ondansetron selectively blocks 5-HT₃ receptors in the chemoreceptor trigger zone and vagal afferents, making it one of the most effective PONV prophylactics with few side effects (mainly QT prolongation at high doses).

Q47. The 'cannot intubate, cannot oxygenate' (CICO) emergency management includes, as a definitive step:

- A) Repeated attempts with the same laryngoscope
- B) Front-of-neck airway (emergency cricothyroidotomy or surgical airway)
- C) Increasing the dose of muscle relaxant
- D) Nasal prongs at high flow

✓ **Answer: B**

Explanation: After failure of facemask ventilation and all supraglottic airway attempts, a front-of-neck airway (emergency cricothyroidotomy) is the final rescue to maintain oxygenation in a CICO scenario, as per Difficult Airway Society guidelines.

Q48. Adrenaline (epinephrine) is the FIRST-LINE treatment for anaphylaxis because it:

- A) Is an antihistamine
- B) Causes alpha-1 vasoconstriction (treating hypotension/angioedema) and beta-2 bronchodilation (treating bronchospasm) and beta-1 positive inotropy
- C) Blocks all histamine receptors
- D) Is a corticosteroid

✓ **Answer: B**

Explanation: Adrenaline simultaneously reverses all life-threatening features of anaphylaxis: restores blood pressure (alpha-1), dilates bronchi (beta-2), and increases cardiac output (beta-1). The dose is 0.5 mg (0.5 mL of 1:1000) IM into the anterolateral thigh.

Q49. The SAFE dose of intrathecal (spinal) hyperbaric bupivacaine 0.5% for a standard adult spinal anaesthetic (e.g., for LSCS or hip arthroplasty) is approximately:

- A) 1–2 mg
- B) 10–15 mg (2–3 mL)
- C) 50 mg
- D) 100 mg

✓ **Answer: B**

Explanation: A typical adult spinal dose of heavy (hyperbaric) bupivacaine 0.5% is 2–3 mL (10–15 mg); 'heavy' solution (specific gravity >1.003) with dextrose sinks in CSF, allowing control of block height by patient positioning.

Q50. Magnesium sulphate is used in obstetric anaesthesia primarily to:

- A) Induce labour
- B) Prevent and treat eclamptic convulsions and as a tocolytic
- C) Reduce postoperative pain only
- D) Reverse neuromuscular blockade

✓ **Answer: B**

Explanation: MgSO₄ is the drug of choice for prophylaxis and treatment of eclampsia (seizures in pre-eclampsia), acting by blocking NMDA receptors and reducing neuromuscular excitability; it also relaxes the uterus (tocolysis).

Q51. Halothane hepatitis, a serious complication, is most common after which pattern of exposure?

- A) First-time exposure in adults
- B) Repeated exposures within a short period, especially in middle-aged obese women
- C) Paediatric use only
- D) Single brief exposure for minor surgery

✓ **Answer: B**

Explanation: Halothane hepatotoxicity (fulminant hepatic necrosis) is an immune-mediated phenomenon linked to TFA (trifluoroacetylated) liver proteins formed on repeated oxidative metabolism; risk is highest with repeated exposures, female sex, obesity, and middle age.

Q52. Which drug is used to treat 'total spinal' block (accidental massive intrathecal local anaesthetic dose with cardiovascular collapse)?

- A) Dantrolene
- B) Vasopressors (e.g., ephedrine, phenylephrine) and supportive care — airway, fluids, atropine for bradycardia
- C) Neostigmine
- D) Naloxone

✓ **Answer: B**

Explanation: Total spinal causes profound sympathetic block, bradycardia, hypotension, and respiratory arrest. Management is entirely supportive: secure the airway (ventilate), give IV fluids and vasopressors to maintain perfusion, and treat bradycardia — the block resolves as the LA wears off.

Q53. Tramadol is classified as a:

- A) Pure opioid agonist (schedule X in India)
- B) Atypical opioid with weak mu-receptor agonism AND noradrenaline/serotonin reuptake inhibition
- C) Non-steroidal anti-inflammatory drug (NSAID)
- D) Local anaesthetic

✓ **Answer: B**

Explanation: Tramadol's dual mechanism — weak mu-receptor agonism plus noradrenaline and serotonin reuptake inhibition — gives it a unique analgesic profile. It can cause seizures and serotonin syndrome, especially in combination with SSRIs.

Q54. Dexmedetomidine produces sedation and analgesia by acting on which receptor?

- A) GABA-A receptor
- B) Alpha-2 adrenergic receptor (central and peripheral)
- C) Opioid mu receptor
- D) NMDA receptor

✓ **Answer: B**

Explanation: Dexmedetomidine is a highly selective alpha-2 agonist (locus coeruleus, spinal cord) producing dose-dependent sedation, analgesia, and anxiolysis without significant respiratory depression — making it useful for awake fiberoptic intubation and ICU sedation.

Q55. Rocuronium at a high dose (1.2 mg/kg, i.e., 3× intubating dose) for RSI can be reversed rapidly with sugammadex at a dose of:

- A) 2 mg/kg
- B) 4 mg/kg
- C) 16 mg/kg
- D) 0.5 mg/kg

✓ **Answer: C**

Explanation: Sugammadex 16 mg/kg ('rescue dose') can reverse even a full RSI dose of rocuronium (1.2 mg/kg) within approximately 3 minutes — an important escape route in a 'cannot intubate' situation where the block needs immediate reversal.

Q56. Which inhalational agent is MOST associated with 'compound A' formation in soda lime, which is nephrotoxic in animal studies?

- A) Isoflurane
- B) Desflurane
- C) Sevoflurane
- D) Halothane

✓ **Answer: C**

Explanation: Sevoflurane reacts with warm soda lime (especially Baralyme) to produce compound A (fluoromethyl-2,2-difluoro-1-[trifluoromethyl]vinyl ether), which is nephrotoxic in rats; minimal human toxicity has been demonstrated, but high-flow circuits and adequate fresh gas flows reduce exposure.

Q57. Bupivacaine is CONTRAINDICATED for Bier's block (IVRA) because:

- A) It does not provide adequate block
- B) Rapid systemic release on cuff deflation can cause severe, often fatal, cardiovascular toxicity
- C) It is too expensive
- D) It causes excessive tourniquet pain

✓ **Answer: B**

Explanation: Bupivacaine IVRA has caused deaths due to sudden release of high drug concentrations on tourniquet deflation, producing refractory ventricular fibrillation. Prilocaine (or lignocaine) is the only agent recommended for IVRA.

Q58. The reversal of benzodiazepine-induced respiratory depression with flumazenil is limited by its:

- A) Long duration (6–8 hours)
- B) Short duration (~45–90 min) — resedation can occur if the offending benzodiazepine has a longer half-life
- C) Inability to cross the blood-brain barrier
- D) Renal toxicity

✓ **Answer: B**

Explanation: Flumazenil's half-life of 45–90 minutes is shorter than most clinical benzodiazepines; patients must be observed for resedation, and repeated doses or a continuous infusion may be required.

Q59. Paracetamol (acetaminophen) IV is particularly useful in multimodal analgesia because it:

- A) Is a potent NSAID that inhibits prostaglandins peripherally
- B) Provides safe central analgesic/antipyretic action without significant platelet inhibition or GI side effects
- C) Causes significant sedation
- D) Reverses neuromuscular blockade

✓ **Answer: B**

Explanation: IV paracetamol provides reliable central analgesia (possibly via endocannabinoid system and spinal serotonergic pathways) without the platelet inhibition (bleeding risk), GI erosion, or renal effects associated with NSAIDs — ideal for day-case and cardiac surgery patients.

Q60. The 'Meyer-Overton hypothesis' of anaesthesia correlates anaesthetic potency with:

- A) Molecular weight
- B) Lipid solubility (oil-gas partition coefficient)
- C) Boiling point of the agent
- D) Saturated vapour pressure

✓ **Answer: B**

Explanation: The Meyer-Overton hypothesis (1899–1901) states that anaesthetic potency correlates with the lipid solubility of the agent — agents that dissolve readily into lipid membranes are more potent (lower MAC). While an oversimplification, this correlation still holds for most clinical agents.

Section D — Patient Monitoring, Fluids & Emergency Management (Q61–80)

Q61. A 'train-of-four' (TOF) stimulation of the ulnar nerve at the wrist is used to assess the DEGREE of:

- A) Depth of inhalational anaesthesia
- B) Neuromuscular blockade (residual muscle relaxant effect)
- C) Opioid analgesia
- D) Regional block spread

✓ **Answer: B**

Explanation: TOF delivers four supramaximal stimuli at 2 Hz; the ratio of 4th to 1st twitch height (TOF ratio) reflects degree of blockade. A TOF ratio ≥ 0.9 is required to confirm adequate recovery of neuromuscular function before extubation.

Q62. Bispectral index (BIS) monitoring is used to measure:

- A) Cardiac output
- B) Depth of hypnosis/anaesthetic effect on the EEG
- C) Neuromuscular function
- D) Blood glucose levels

✓ **Answer: B**

Explanation: The BIS monitor analyses frontal EEG signals and produces a dimensionless number (0–100); a BIS of 40–60 targets appropriate general anaesthesia depth, reducing risk of awareness and of drug overdose.

Q63. Pulse pressure variation (PPV) and stroke volume variation (SVV) are dynamic preload indicators used in mechanically ventilated patients. A PPV >13% suggests:

- A) Fluid overload
- B) Fluid responsiveness — the patient is likely to increase cardiac output in response to a fluid bolus
- C) Cardiac tamponade
- D) Pulmonary oedema

✓ **Answer: B**

Explanation: In mechanically ventilated patients without arrhythmias, PPV >13% indicates significant cyclic variation in preload from positive-pressure breathing, predicting that cardiac output will rise with fluid administration ('fluid-responsive').

Q64. In a massive transfusion scenario, fresh frozen plasma (FFP) is indicated to correct:

- A) Anaemia only
- B) Coagulation factor deficiencies (replaces clotting factors depleted by dilution/consumption)
- C) Thrombocytopenia only
- D) Hyperkalaemia

✓ **Answer: B**

Explanation: FFP contains all coagulation factors; in massive haemorrhage, clotting factors are diluted by crystalloid resuscitation and consumed in DIC. FFP (and cryoprecipitate for fibrinogen) corrects the coagulopathy that perpetuates bleeding.

Q65. The 'Parkland formula' for fluid resuscitation of burns calculates the volume as:

- A) $2 \text{ mL} \times \text{weight (kg)} \times \% \text{ BSA burn}$
- B) $4 \text{ mL} \times \text{weight (kg)} \times \% \text{ BSA burn}$ of Ringer Lactate in 24 hours
- C) 10 mL/kg of colloid over 12 hours
- D) 1 L/hr of normal saline for all adults

✓ **Answer: B**

Explanation: The Parkland (Baxter) formula estimates $4 \text{ mL} \times \text{kg} \times \% \text{BSA burned}$ of Ringer Lactate in the first 24 hours; half is given in the first 8 hours (from time of burn) and the remainder over 16 hours — used for burns >20–25% BSA.

Q66. During CPR, adrenaline (epinephrine) 1 mg IV is given every:

- A) 30 seconds
- B) 2 minutes (every other CPR cycle)
- C) 3–5 minutes
- D) 10 minutes

✓ **Answer: C**

Explanation: Current ALS guidelines recommend adrenaline 1 mg IV every 3–5 minutes throughout cardiac arrest. For VF/pVT, it is given after the 3rd shock; for non-shockable rhythms (PEA/asystole), it is given as soon as IV access is established.

Q67. Amiodarone 300 mg IV is given during cardiac arrest in the context of:

- A) Asystole
- B) Pulseless electrical activity (PEA)
- C) Shock-refractory ventricular fibrillation after the 3rd defibrillation attempt
- D) Sinus bradycardia

✓ **Answer: C**

Explanation: ALS protocols recommend amiodarone 300 mg IV after the 3rd defibrillation attempt in refractory VF/pVT, with a second dose of 150 mg after the 5th shock; it stabilises ventricular rhythm by multiple ion-channel effects.

Q68. A reading of SpO₂ 97% on pulse oximetry in a patient receiving supplemental oxygen indicates:

- A) Significant hypoventilation
- B) Adequate peripheral oxygen saturation, but does not reflect ventilation (PaCO₂ may be rising)
- C) Carbon monoxide poisoning excluded
- D) No anaemia present

✓ **Answer: B**

Explanation: Pulse oximetry measures only oxygen saturation — it cannot detect hypercarbia (rising CO₂) or carbon monoxide poisoning (COHb has the same absorbance as OHb at these wavelengths); ABG is needed for a full respiratory picture.

Q69. The Mallampati Score is assessed with the patient:

- A) Supine with neck extended
- B) Sitting upright, mouth wide open, tongue maximally protruded, without phonating
- C) Lying laterally
- D) Under anaesthesia

✓ **Answer: B**

Explanation: Standardised positioning (seated, mouth maximally open, tongue protruded, no phonation) is essential for reproducibility; phonation elevates the soft palate and overestimates airway access. The visible structures (faucial pillars, soft palate, uvula) determine the class.

Q70. Intraoperative hypothermia (core temperature <36°C) is associated with all of the following EXCEPT:

- A) Increased surgical site infection rate
- B) Prolonged drug metabolism (slower recovery from anaesthesia)
- C) Reduced intraoperative blood loss (protective coagulation effect)
- D) Coagulopathy and increased transfusion requirement

✓ **Answer: C**

Explanation: Hypothermia IMPAIRS coagulation (reduces platelet function and clotting factor activity), INCREASING blood loss and transfusion requirements — not reducing it. It also increases SSI risk and prolongs drug metabolism.

Q71. In paediatric cardiac arrest, the INITIAL defibrillation energy dose is:

- A) 50 J fixed
- B) 200 J fixed
- C) 4 J/kg
- D) 360 J fixed

✓ **Answer: C**

Explanation: Paediatric defibrillation uses weight-based dosing: 4 J/kg for all shocks (ALS guidelines 2021); adult ALS uses 200 J (biphasic) for the first shock regardless of weight.

Q72. The 'Rule of Nines' for estimating body surface area (BSA) in burn assessment allocates what percentage to the entire head and neck in an ADULT?

- A) 18%
- B) 9%
- C) 4.5%
- D) 1%

✓ **Answer: B**

Explanation: In adults: head+neck = 9%, each upper limb = 9%, anterior trunk = 18%, posterior trunk = 18%, each lower limb = 18%, perineum = 1%. In infants/young children, the head is proportionally larger (18%), corrected by Lund-Browder charts.

Q73. Forced-air warming blankets placed over a patient during surgery are used to actively prevent:

- A) Surgical site infection via heat sterilisation of the skin
- B) Perioperative hypothermia by convective heat transfer
- C) Excess diuresis
- D) Blood pressure fluctuation

✓ **Answer: B**

Explanation: Forced-air warming blankets deliver warm air across the patient's body surface via convection — the most effective and commonly used method of intraoperative warming, significantly reducing the incidence of perioperative hypothermia.

Q74. Soda lime turns which colour when EXHAUSTED and no longer absorbs CO₂ effectively?

- A) Red
- B) Blue/purple (indicator dye change, e.g., ethyl violet)
- C) Yellow
- D) Green

✓ **Answer: B**

Explanation: Soda lime contains pH-sensitive indicator dyes (commonly ethyl violet) that turn blue/purple as the CO₂-absorbing capacity is exhausted and pH falls; some formulations use other colours — but the colour change is always a warning to replace the absorbent.

Q75. The 'Trendelenburg test' (passive leg raise, PLR) as a dynamic fluid responsiveness assessment works by:

- A) Assessing haemoglobin level change
- B) Autotransfusing ~300 mL from the lower limbs to the thorax and measuring the cardiac output response
- C) Measuring SpO₂ change with position
- D) Quantifying urine output change

✓ **Answer: B**

Explanation: PLR tilts the patient to 45° from semi-recumbent, transferring blood from the lower limbs to the central circulation ('autotransfusion' of ~300 mL); a rise in cardiac output (measured by non-invasive CO monitor or arterial waveform) confirms preload-responsiveness.

Q76. Blood warming during intraoperative transfusion is important primarily to prevent:

- A) Haemolysis from high temperature
- B) Hypothermia, hypocalcaemia, and cardiac arrhythmias from rapid infusion of cold blood products
- C) Bacterial contamination
- D) Coagulation activation

✓ **Answer: B**

Explanation: Blood stored at 4°C, if infused rapidly without warming, can cause core hypothermia, worsen coagulopathy, cause citrate-induced hypocalcaemia, and precipitate cardiac arrhythmias. In-line blood warmers (e.g., Level 1, Ranger) heat blood to ~37°C.

Q77. The APGAR score of 7–10 at 1 minute indicates:

- A) Immediate need for resuscitation
- B) Normal neonate — routine care
- C) Moderate depression requiring stimulation and oxygen
- D) Severe depression requiring intubation

✓ **Answer: B**

Explanation: APGAR 7–10 = normal/healthy neonate (routine care); 4–6 = moderate depression (stimulation, supplemental O₂, positive pressure ventilation if not improving); 0–3 = severe depression (immediate resuscitation: PPV, chest compressions, drugs).

Q78. A patient in the PACU (recovery room) develops stridor. The most common cause in the first 30 minutes post-extubation is:

- A) Pulmonary embolism
- B) Laryngospasm or post-extubation glottic oedema/secretions
- C) Pneumothorax
- D) Myocardial infarction

✓ **Answer: B**

Explanation: Stridor in early recovery is most commonly laryngospasm (high-pitched, responds to jaw thrust and CPAP/suxamethonium if severe) or glottic oedema (from prolonged intubation or difficult laryngoscopy); heliox or nebulised adrenaline may help oedema.

Q79. The CHCT (Caffeine Halothane Contracture Test) is the gold-standard diagnostic test for:

- A) Difficult intubation
- B) Malignant hyperthermia susceptibility (MHS)
- C) Local anaesthetic allergy
- D) Pseudocholinesterase deficiency

✓ **Answer: B**

Explanation: The CHCT (or IVCT in Europe) exposes muscle biopsy strips to caffeine and halothane; abnormal contracture response indicates MHS — the only definitive test because genetic testing alone misses ~20–30% of susceptible individuals.

Q80. The 'difficult airway trolley' must be immediately accessible in all areas where anaesthesia or deep sedation is administered. Which of the following is NOT typically found on it?

- A) Video laryngoscope
- B) Supraglottic airway devices (ProSeal, i-gel)
- C) Defibrillator paddles
- D) Surgical airway kit (cricothyroidotomy set)

✓ **Answer: C**

Explanation: A difficult airway cart stocks airway rescue equipment: video/optical laryngoscopes, various LMAs/SGAs, bougies, fibreoptic scope (or bronchoscope), jet ventilation kit, and a surgical airway kit. A defibrillator is kept on the crash trolley, not the airway cart.

Section E — Surgical Procedures, Positions & Perioperative Care (Q81–100)

Q81. The 'park bench' (lateral oblique) position is specifically used for:

- A) Gynaecological laparoscopy
- B) Posterior fossa and cerebello-pontine angle neurosurgery
- C) Appendectomy
- D) Cardiac surgery via median sternotomy

✓ **Answer: B**

Explanation: The park bench position — lateral with trunk tilted slightly prone, head in Mayfield pins/horseshoe — gives optimal access to the posterior cranial fossa and cerebellopontine angle for acoustic neuroma and other posterior fossa surgery.

Q82. The 'beach chair' (semi-sitting) position used for shoulder arthroscopy creates anaesthetic risk of:

- A) Lower limb nerve compression
- B) Venous air embolism and cerebral hypoperfusion (head-up reduces cerebral perfusion pressure)
- C) Abdominal compartment syndrome
- D) Retinal ischaemia from prone position

✓ **Answer: B**

Explanation: With the head elevated, the gradient from heart to brain reduces cerebral perfusion pressure; in addition, veins in the surgical field are below atmospheric pressure, creating risk of VAE. MAP at heart level may under-represent brain perfusion — many centres target MAP at ear level.

Q83. Pfannenstiel incision, a transverse suprapubic incision through which the rectus sheath is divided, is most commonly used for:

- A) Appendectomy
- B) Caesarean section and pelvic gynaecological surgery
- C) Cholecystectomy
- D) Renal transplant

✓ **Answer: B**

Explanation: The Pfannenstiel incision provides excellent cosmesis (hidden in the pubic hairline) and good pelvic access; it is the standard for caesarean section, myomectomy, and other open pelvic procedures.

Q84. In laparoscopic surgery, the 'Hasson technique' (open entry) is preferred over the Veress needle technique primarily in patients with:

- A) Obesity
- B) Previous abdominal surgery (risk of adhesions) or suspected haemoperitoneum
- C) Pregnancy beyond 1st trimester
- D) All of the above

✓ **Answer: D**

Explanation: All three situations increase the risk of injury with blind Veress needle insertion — adhesions from prior surgery, distorted anatomy in pregnancy, and potential for injury to dilated vessels in haemoperitoneum. Hasson's open (cut-down) technique allows direct visualisation before inserting the port.

Q85. The nerve most at risk during a standard inguinal hernia repair (open mesh/Lichtenstein) is the:

- A) Femoral nerve
- B) Ilioinguinal nerve (and genitofemoral nerve)
- C) Sciatic nerve
- D) Lateral femoral cutaneous nerve

✓ **Answer: B**

Explanation: The ilioinguinal nerve runs within the inguinal canal along the cord; damage during dissection or mesh fixation causes chronic groin pain, medial thigh hypoaesthesia, and scrotal numbness — the most common neuropathic complication of open inguinal hernia repair.

Q86. The 'Whipple procedure' (pancreaticoduodenectomy) involves resection of:

- A) The spleen and tail of pancreas
- B) The head of pancreas, duodenum, distal common bile duct, and gallbladder — with reconstruction
- C) The stomach and oesophagus
- D) The sigmoid colon only

✓ **Answer: B**

Explanation: The Whipple procedure removes the head of pancreas (where most pancreatic cancers arise), duodenum, gallbladder, and distal CBD, followed by complex reconstruction (pancreaticojejunostomy, hepaticojejunostomy, gastrojejunostomy).

Q87. A 'tourniquet' applied to a limb for bloodless field surgery should ideally not be inflated continuously for more than:

- A) 15 minutes
- B) 30 minutes
- C) 1.5–2 hours (with warning at 90 minutes)
- D) 4 hours

✓ **Answer: C**

Explanation: Standard practice limits tourniquet inflation to <2 hours to avoid ischaemia-reperfusion injury, nerve palsy, and rhabdomyolysis; a warning should be given to the surgeon at 90 minutes. The tourniquet is typically inflated to systolic pressure + 50–75 mmHg for the lower limb and + 50 mmHg for the upper limb.

Q88. Thyroidectomy carries a specific risk of which immediate postoperative complication due to inadvertent removal of the parathyroid glands?

- A) Hyperglycaemia
- B) Hypocalcaemia (tetany, stridor, seizures)
- C) Hyperthermia
- D) Pulmonary embolism

✓ **Answer: B**

Explanation: Parathyroid glands regulate calcium; their devascularisation or inadvertent excision causes hypoparathyroidism with hypocalcaemia, manifesting as perioral and digital tingling (Chvostek's/Trousseau's signs), muscle cramps, tetany, laryngospasm, and seizures.

Q89. In cardiac surgery requiring cardiopulmonary bypass (CPB), the pump oxygenator takes over the function of:

- A) The kidneys only
- B) Both the heart (pumping) and the lungs (oxygenation/CO₂ removal), allowing the surgeon to operate on a still, bloodless heart
- C) The liver
- D) The brain

✓ **Answer: B**

Explanation: CPB diverts systemic venous blood through an oxygenator (gas exchange) and roller/centrifugal pump (circulation), completely substituting cardiopulmonary function during open-heart surgery while the aorta is cross-clamped and the heart is arrested with cardioplegia.

Q90. Myringotomy (tympanotomy) — incision of the tympanic membrane — is a common paediatric ENT procedure. The anaesthetic of choice (preferred to avoid airway manipulation) is usually:

- A) Spinal anaesthesia
- B) Inhalational anaesthesia maintained via facemask (sevoflurane), avoiding intubation
- C) Epidural anaesthesia
- D) Regional nerve block only

✓ **Answer: B**

Explanation: Myringotomy is extremely brief (2–3 minutes per ear); facemask sevoflurane anaesthesia allows adequate depth without intubation, and the child recovers almost immediately, facilitating a high-volume day-case service.

Q91. A 'Hartmann's procedure' involves:

- A) Resection of the sigmoid colon with end colostomy and closure of the rectal stump (without anastomosis)
- B) Formation of a loop ileostomy only
- C) Right hemicolectomy with primary anastomosis
- D) Colostomy reversal

✓ **Answer: A**

Explanation: Hartmann's procedure is performed for perforated sigmoid diverticulitis, obstructing colorectal cancer, or trauma; the diseased segment is resected, the proximal end brought out as an end colostomy, and the rectal stump closed — avoiding a high-risk anastomosis in a contaminated/unstable patient.

Q92. Nephrectomy via a flank (loin) incision most commonly places the patient in the:

- A) Supine position
- B) Lateral (kidney) position with the table flexed and kidney rest elevated
- C) Prone position
- D) Lithotomy position

✓ **Answer: B**

Explanation: The lateral position with table flexion and kidney rest opens the flank by spreading the costal margin from the iliac crest, directly exposing the retroperitoneum; care must be taken with the axillary roll and lower-limb padding to prevent pressure injuries.

Q93. In ophthalmology, the 'oculocardiac reflex' (OCR) triggered by traction on the extraocular muscles or pressure on the globe causes:

- A) Hypertension and tachycardia
- B) Bradycardia, nodal rhythm, or even asystole via the trigemino-vagal arc
- C) Mydriasis
- D) Nystagmus

✓ **Answer: B**

Explanation: OCR is mediated by afferent (ophthalmic division of V1 → ciliary ganglion) and efferent (vagus nerve → SA node) pathways; bradycardia can be severe and requires the surgeon to release traction, and IV atropine if persistent.

Q94. The modified radical mastectomy (MRM) differs from a simple (total) mastectomy in that MRM additionally includes:

- A) Removal of the pectoralis major muscle
- B) Axillary lymph node dissection (levels I and II)
- C) Chest wall resection
- D) Omentoplasty

✓ **Answer: B**

Explanation: Simple mastectomy removes the breast only; MRM removes the breast plus levels I and II axillary lymph nodes (without removing the pectoralis muscles as in the older Halsted radical mastectomy), allowing staging and reducing axillary recurrence.

Q95. The term 'anastomotic leak' in colorectal surgery refers to:

- A) Haemorrhage from the anastomosis site
- B) Failure of healing of the intestinal join, allowing faecal contamination and peritonitis
- C) Obstruction at the anastomosis due to stricture
- D) Wound dehiscence

✓ **Answer: B**

Explanation: Anastomotic leak (typically presenting day 3–7 post-op with fever, peritonism, and rising inflammatory markers) is one of the most feared complications of colorectal surgery, often requiring return to theatre for re-resection or defunctioning stoma.

Q96. In urological surgery, transurethral resection of the prostate (TURP) is performed using a resectoscope with an electrosurgical loop. Spinal anaesthesia is preferred over GA for TURP because it:

- A) Provides better muscle relaxation for the abdomen
- B) Allows early neurological signs of TURP syndrome (confusion, visual disturbance) to be detected in the awake patient
- C) Reduces tourniquet pain
- D) Is required for the lithotomy position

✓ **Answer: B**

Explanation: With spinal anaesthesia, the patient is awake and the early CNS features of TURP syndrome (hyponatraemia/glycine toxicity: confusion, nausea, visual disturbances) are detectable before cardiovascular collapse occurs, allowing prompt intervention.

Q97. Robotic-assisted surgery (e.g., Da Vinci system) presents which UNIQUE anaesthetic challenge compared to standard laparoscopy?

- A) Lower pneumoperitoneum pressure needed
- B) Robot arms dock to patient — the anaesthetist CANNOT access the patient easily once docked; all lines/airway must be secured beforehand
- C) General anaesthesia is never required
- D) No Trendelenburg position needed

✓ **Answer: B**

Explanation: Once the robotic arms are docked to the patient's ports, access to the patient's airway, IV lines, and body is severely restricted. All positioning (often steep Trendelenburg), lines, and airway management must be completed and secured before docking.

Q98. Compartment syndrome, a surgical emergency, is characterised by the '6 Ps'. Which of the following is the EARLIEST and most sensitive symptom?

- A) Paralysis
- B) Pulselessness
- C) Pain out of proportion to the injury, worsened by passive stretch of the compartment muscles
- D) Pallor

✓ **Answer: C**

Explanation: Pain disproportionate to the injury — especially severe pain on passive muscle stretch — is the earliest warning sign of compartment syndrome (raised interstitial pressure impairing perfusion); paralysis and pulselessness are late, ominous signs. Treatment is emergency fasciotomy.

Q99. Laparoscopic cholecystectomy is performed through how many port sites in the standard 4-port technique?

- A) 1
- B) 2
- C) 3
- D) 4

✓ **Answer: D**

Explanation: The standard technique uses 4 ports: one 10–12 mm umbilical port (camera), one 10 mm epigastric port (clips/retractor), and two 5 mm ports in the right upper quadrant for grasping and dissection. Single-incision laparoscopic (SILS) variants use 1 port.

Q100. The WHO Surgical Safety Checklist has three distinct pause points. Which list is completed BEFORE the patient leaves the operating room?

- A) Sign In (before anaesthesia induction)
- B) Time Out (before skin incision)
- C) Sign Out (before the patient leaves the OT)
- D) Transfer Out (in the recovery room)

✓ **Answer: C**

Explanation: The WHO checklist Sign Out phase confirms: instrument/sponge/needle count correct, specimen labelling, any equipment concerns, and key recovery/management information — performed before the patient is taken from the OR.

Quick-Reference Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	21	B	41	D	61	B	81	B
2	C	22	B	42	B	62	B	82	B
3	B	23	C	43	B	63	B	83	B
4	B	24	B	44	B	64	B	84	D
5	C	25	B	45	B	65	B	85	B
6	C	26	B	46	B	66	C	86	B
7	B	27	B	47	B	67	C	87	C
8	B	28	B	48	B	68	B	88	B
9	B	29	B	49	B	69	B	89	B
10	B	30	C	50	B	70	C	90	B
11	B	31	B	51	B	71	C	91	A
12	C	32	B	52	B	72	B	92	B
13	B	33	D	53	B	73	B	93	B
14	B	34	B	54	B	74	B	94	B
15	C	35	B	55	C	75	B	95	B
16	B	36	B	56	C	76	B	96	B
17	A	37	B	57	B	77	B	97	B
18	B	38	B	58	B	78	B	98	C
19	B	39	A	59	B	79	B	99	D
20	B	40	C	60	B	80	C	100	C

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